Traditional ML

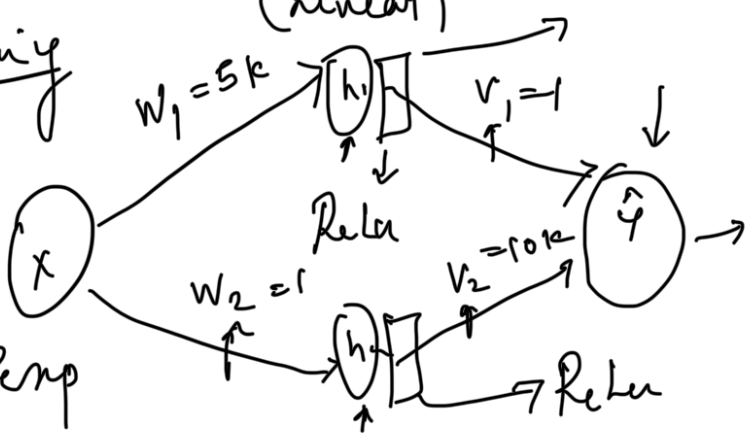
↳ Subscribe & Stay true to the family of their implementation

L.R → Line Eqn

Logistic Regression

(Linear)

Deeplearning



Exp

Neuron 1

$(5x + 20) = z \rightarrow \text{Relu}^{\max}(0, z)$

$\max(0, 5x+20)$

Linear Regression?

Yes

↓

Passed

Neuron 2

$z = (x-3) \rightarrow \text{Relu}^{\max}(0, x-3)$

Non-Linearity

Neural Networks understand

Global patterns in my data.

$\hat{y} = 1 \times \text{Relu}(5k x_1 + 20) + 10k \text{ Relu}(x-3)$

No. of Neurons /

$$\hat{y} = V_1 h_1 + V_2 h_2 + C$$

$$h_1 = \text{Relu}(w_1 x + b_1)$$

$$h_2 = \text{Relu}[w_2 x + b_2)$$

No. of hidden / Layer, choice of Non-linearity

HYPERPARAMS

x = 5 years = 7

N_1

$h_1 = 4(5) + 20 = 40 \rightarrow \text{Relu} = 40 \rightarrow$

$h_2 = 1(5) - 3 = 2 \rightarrow \text{Relu} = 2$ $N_2 \rightarrow$

$$\hat{y} = 1(40) + 10(2) + 0$$

$$\hat{y} = 60k$$

FORWARD PASS! ⇒ How well has my NN learnt.

Actual → 65k

Loss! $L = \frac{1}{2}(y - \hat{y})^2$

$$L = \frac{1}{2}(60 - 65)^2 = \frac{1}{2}(-5)^2 = 12.5k$$

BACKPROPAGATION

Two directions of info flow.

i) forward prop

I/P → Hidden → Output → Loss ✓

ii) Backward prop.

I/p side ← Hidden layer gradients ← output gradients ← loss

gradients :

How does my loss change when I make a small change to a particular weight, while holding other parms fixed.

Differentiation → $\frac{dL}{dW}$

+ve gradient → ↑ parm ↑ loss

-ve gradien → ↑ parm ↓ loss

$L = \frac{1}{2}(\hat{y} - y)^2$ — ① Is differentiable ?

$\frac{dL}{d\hat{y}} = \hat{y} - y = -5$

60 65

The model learns by parameter update using ① loss

② Metrics

Are human understandable quantification of Model perf

Classification

$= -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$

Acc ⇒ $\frac{TP+TN}{TP+TN+FP+FN}$

$\frac{dAcc}{dW}$ → That is not differentiable cannot go through

Output Neuron

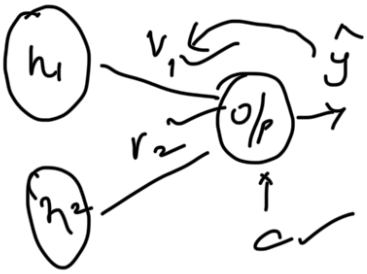
$\hat{y} = v_1 h_1 + v_2 h_2 + C$

$h_1 = 40 \mid h_2 = 2$

$v_1 = 1 \mid v_2 = 10$

$\frac{dL}{dv_1} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dv_1} \quad \Big| \quad \frac{d\hat{y}}{dv_1} = h_1 = 40$

Q.D



$\therefore \quad \frac{dL}{dv_1} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dv_1} = -5(40) = -200$

$$\rightarrow \frac{\partial L}{\partial v_1}$$

$$\frac{\partial L}{\partial r_2} = (-5)\,2 = -10$$

$$\rightarrow \frac{\partial L}{\partial r_2}$$

$$\frac{\partial L}{\partial c} = -5$$

$$\rightarrow \frac{\partial L}{\partial c}$$

---

first hiddenlayer.

$$\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial b}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_1} \cdot \frac{\partial h_1}{\partial z} \cdot \frac{\partial z_1}{\partial w_1} \quad (1)$$

How does my loss change w.r.t Predicted Salary

How does my Salary change with this hidden neuron's o/p

How does my hidden neuron o/p change w.r.t weighted sum

How much does my weighted sum change w.r.t change in w_1

Backpropagation Measures contributions of each param. towards my loss.

Parameters	Current value	Gradients
w_1	4	-25
b_1	20	-5
w_2	1	-250
b_2	-3	-50
v_1	1	-200

v_2 c. learning rate $\eta = 0.0001$ | 10 | -5

$$\theta_{new} = \theta_{old} - (\eta / \lambda)\, dH/d\theta \quad \text{↑ L:R}$$

$w_1' = 4 - 0.0001(-25) = 4.0025$

$w_2' = 1 - 0.0001(-250) = 1.025$

Parameter	Before	After
w_1	4	4.0025
b_1	20	20.005
w_2	1	1.025
b_2	-3	-2.995
v_1	1	1.0200
v_2	10	10.0010
c	0	0.0005

$h_1 = \text{Relu}(4.0025(5) + 20.005) = 40.013$

$h_2 = \text{Relu}(1.025(5) - 2.995) = 2.13$

$\hat{y} = 1.02(40.013) + 10.001(2.13) + 0.0005$

≈ 62.16

Prev 60k → Current 62.16k → Actual 65k // MSE

over 100 rows

1 Iteration → 1 Batch = 100 rows

I/P DATA → FORWARD PASS → LOSS CALCULATION → BACK PROP → OPTIMIZER STEP

EPOCH | BATCH | ITERATION

EPOCH - 1 complete pass through the entire Dataset

1000 rows → forward pass → o/ps

← Change w's & b's

Batch

1000 → 100 record Batch

1000 rows

Small Batch (100) ... 100 ... 100

100 rows → Forward pass → loss

← Backprop

Change w's & B's

I finish 1 epoch in 10 Iterations.

Iteration:

One optimizer step. Here one Opt. step for each Batch.

10 Iteration → 1 epoch

⇒ Per row ⇒ Sensitive optimizer, each row pattern is taken account

SE = $\frac{1}{2}(\hat{y}-y)^2$

⇒ Little Smoothed ⇒ 10 row / 100 rows / 1000 rows

MSE = $\frac{1}{2N}(\hat{y}-y)^2$

⇒ full smooth ⇒ per epoch Backprop

MSE = $\frac{1}{2N}(\hat{y}-y)^2$